

Backend Developer Roadmap (Placement-Focused)

This document is a **complete, step-by-step backend learning guide** designed specifically for **placements and internships**. It focuses on **Node.js backend development using TypeScript and MongoDB**, which is widely used in the industry.

1. Goal of This Roadmap

- Become a **job-ready Backend Developer**
 - Build **real backend projects** for resume & GitHub
 - Be confident in **backend interview questions**
 - Use **TypeScript (industry standard)** instead of only JavaScript
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2. Final Tech Stack You Will Use

- JavaScript (Core fundamentals)
 - TypeScript (Professional backend development)
 - Node.js
 - Express.js
 - MongoDB
 - Mongoose
 - REST APIs
 - JWT Authentication
 - Git & GitHub
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3. JavaScript Fundamentals (Before TypeScript)

You must first understand **JavaScript basics**, because TypeScript is built on top of JavaScript.

Topics to Learn

- Variables: let, const, var
- Data types
- Operators
- Conditional statements
- Loops
- Functions (normal & arrow functions)
- Objects and Arrays
- Destructuring
- Spread & Rest operators
- Array methods: map, filter, reduce, find

YouTube Resources

- **CodeWithHarry – JavaScript Complete Course**
- **Namaste JavaScript – Akshay Saini**

Focus especially on understanding **how JavaScript works internally**.

4. Asynchronous JavaScript (Very Important)

Backend development depends heavily on async operations.

Topics to Learn

- Callbacks
- Promises
- `async / await`
- Error handling using `try/catch`
- `setTimeout` and event loop basics

YouTube Resources

- **Namaste JavaScript – Akshay Saini (Async JS episodes)**
 - **Web Dev Simplified – Async JavaScript & Error Handling**
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5. ES6+ JavaScript Features

These features are used heavily in TypeScript and modern backend code.

Topics to Learn

- Arrow functions
- Template literals
- Modules (`import / export`)
- Optional chaining
- Nullish coalescing

YouTube Resources

- **Traversy Media – Modern JavaScript (ES6+)**
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6. Switching to TypeScript

Once you are comfortable with JavaScript fundamentals, you should **switch to TypeScript**.

Why TypeScript?

- Fewer runtime errors

- Better code readability
- Industry-preferred for backend
- Strong typing & scalability

TypeScript Topics to Learn

- Basic types
- Type inference
- Interfaces
- Type aliases
- Enums
- Generics
- tsconfig

YouTube Resources

- **CodeWithHarry – TypeScript Tutorial**
 - **Traversy Media – Node.js with TypeScript Setup**
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7. Node.js Fundamentals

Node.js allows JavaScript/TypeScript to run on the server.

Topics to Learn

- What is Node.js
- Event loop
- fs module
- Environment variables
- Project structure

YouTube Resources

- **CodeWithHarry – Node.js Backend Tutorial**
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8. Express.js (Backend Core)

Express.js is the framework used to build REST APIs.

Topics to Learn

- REST API concepts
- Routing
- Middleware
- Request & Response objects
- Error handling
- MVC architecture

YouTube Resources

- Traversy Media – Express.js Crash Course
 - Web Dev Simplified – Express with TypeScript
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9. MongoDB & Mongoose

MongoDB is a NoSQL database commonly used with Node.js.

Topics to Learn

- CRUD operations
- Schema design
- Mongoose models
- Relationships
- Validation
- Basic aggregation

YouTube Resources

- The Net Ninja – MongoDB Tutorial
 - The Net Ninja – Mongoose Tutorial (with TypeScript)
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10. Authentication & Security (Placement Critical)

Almost every backend interview asks about authentication.

Topics to Learn

- JWT authentication
- Password hashing (bcrypt)
- Login & Register APIs
- Protected routes
- Role-based access control

YouTube Resources

- Web Dev Simplified – JWT Authentication in Node.js
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11. Backend Projects (Mandatory)

Projects are essential for placements.

Recommended Projects

1. Authentication System (JWT + TypeScript)
2. Blog API (CRUD + Auth)

3. Task Management API

YouTube Resource

- [Amigoscode – Node.js TypeScript REST API \(Full Project\)](#)
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12. Git & GitHub

Recruiters will check your GitHub profile.

Topics to Learn

- Git basics
- Commits & branches
- README writing
- .env handling

YouTube Resource

- [CodeWithHarry – Git & GitHub Tutorial](#)
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13. Interview Preparation (Backend)

Topics

- REST vs GraphQL
- HTTP status codes
- MVC architecture
- JWT working
- Error handling strategies

YouTube Resource

- [Fireship – Backend & System Design Basics](#)
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14. How to Mention Skills on Resume

Languages: JavaScript (ES6+), TypeScript Backend: Node.js, Express.js Database: MongoDB, Mongoose Auth: JWT, bcrypt Tools: Git, GitHub, Postman

15. Final Outcome

After completing this roadmap, you will: - Have **strong JavaScript fundamentals** - Build backend systems using **TypeScript** - Be confident for **backend placements & internships** - Have real projects on GitHub

16. Day-1 to Day-60 Timetable (Based on College Schedule)

College Schedule: 7:00 AM – 7:00 PM

Daily Fixed Time Blocks (Detailed)

- **6:00 – 6:10 AM** → Wake up & light refresh
- **6:10 – 6:45 AM** → Backend learning (JS / TS / Node concepts)
- **7:00 AM – 7:00 PM** → College + travel
- **7:00 – 8:00 PM** → Dinner + rest (NO coding)
- **8:00 – 8:45 PM** → **DSA practice (core placement topics)**
- **9:00 – 10:30 PM** → Backend coding & implementation
- **10:30 – 11:00 PM** → Debugging, revision & GitHub commit



Day 1 – Day 7: JavaScript Fundamentals

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Variables, data types, operators - Conditions and loops - Functions (normal & arrow)



Resource: - CodeWithHarry – JavaScript Complete Course



Day 8 – Day 14: Objects, Arrays & ES6

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Objects & arrays - Destructuring - Spread & rest operators - map, filter, reduce - ES6 features



Resource: - Namaste JavaScript – Akshay Saini - Traversy Media – Modern JavaScript



Day 15 – Day 21: Asynchronous JavaScript

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Callbacks - Promises - async / await - try/catch - Event loop basics



Resource: - Namaste JavaScript (Async JS episodes)



Day 22 – Day 27: TypeScript Basics

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Types - Interfaces - Type inference - Enums - Generics



Resource: - CodeWithHarry – TypeScript Tutorial

Day 28 – Day 32: Node.js Fundamentals

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Node.js intro - Event loop - fs module - Environment variables - Project structure

 Resource: - CodeWithHarry – Node.js Backend Tutorial

Day 33 – Day 38: Express.js + REST APIs

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - Express setup - Routing - Middleware - MVC structure - Error handling

 Resource: - Traversy Media – Express Crash Course - Web Dev Simplified – Express with TypeScript

Day 39 – Day 45: MongoDB & Mongoose

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - CRUD operations - Schema design - Models - Relations

 Resource: - The Net Ninja – MongoDB & Mongoose

Day 46 – Day 50: Authentication & Security

Daily Timing: 6:00–6:45 AM (Learning) | 9:00–11:00 PM (Practice) - JWT - bcrypt - Login/Register APIs - Protected routes

 Resource: - Web Dev Simplified – JWT Authentication

Day 51 – Day 57: Backend Projects

Daily Timing: 6:00–6:45 AM (Planning/Review) | 9:00–11:00 PM (Implementation) - Auth System - Blog API or Task Manager API - GitHub push + README

 Resource: - Amigoscode – Node.js TypeScript REST API

Day 58 – Day 60: Revision & Interview Prep

Daily Timing: 6:00–6:45 AM (Revision) | 9:00–11:00 PM (Mock Q&A / Practice) - Revise core concepts - Practice explaining projects - Backend interview questions

 Resource: - Fireship – Backend Concepts

17. DSA Plan (Alongside Backend)

DSA is included in this roadmap **only for placement-level preparation**, without affecting backend focus.

Daily DSA Time

- **8:00 – 8:45 PM (45 minutes)**
- Practice **5–6 days per week**

Core DSA Topics for Placements (Must-Do)

These topics are **sufficient for most service + product company placements**. Focus on **understanding patterns**, not memorizing solutions.

Tier-1 (HIGH PRIORITY – Must Finish)

1. Arrays
2. Prefix sum
3. Two pointers
4. Sliding window
5. Strings
6. Frequency counting
7. Palindrome problems
8. Hashing (HashMap / HashSet)
9. Counting problems
10. Subarray problems
11. Stack
12. Next greater element
13. Balanced parentheses
14. Queue
15. Basic queue & deque usage
16. Linked List
17. Reverse linked list
18. Detect cycle
19. Fast & slow pointers

Tier-2 (MEDIUM PRIORITY – Good to Know)

1. Binary Search
2. Search in sorted array
3. First & last occurrence
4. Sorting
5. Selection, Bubble (concept)
6. Merge sort (understanding)
7. Recursion (Basic)
8. Subsets
9. Simple backtracking logic

Tier-3 (LOW PRIORITY – Optional for Now)

1. Trees (Basics only)

2. Binary tree traversal (inorder, preorder, postorder)

You can SKIP for now (unless targeting top product companies): - Advanced Dynamic Programming - Graphs - Tries - Segment Trees

DSA Resources

- **Striver – A2Z DSA Course (YouTube)**
- **LeetCode** (Easy → Medium)
- **GeeksforGeeks** (topic-wise practice)

30-Day DSA Side Plan

- Days 1–4: Arrays
- Days 5–7: Strings
- Days 8–10: Recursion
- Days 11–14: Linked List
- Days 15–18: Stack & Queue
- Days 19–21: Hashing
- Days 22–25: Searching & Sorting
- Days 26–30: Mixed practice

18. Next Recommended Steps

- Practice API testing with Postman
- Deploy projects (Render / Railway)
- Apply for Backend Intern & Node.js roles

End of Document